# Data, Information, Knowledge

## Data

Data are the raw facts, observations, or recorded values. Data by itself is often meaningless or unstructured. Data are just symbols, numbers, text, or records. Data by itself is meaningless and has no context.

## Information

When data is organized, structured, processed and given context, it becomes information. Information answers questions like *who*, *what*, *when*, *where* giving meaning to raw data.

## Knowledge

Knowledge is build when information is interpreted, analyzed, or connected with experience to form understanding. It answers "how" and "why" questions, and can be used for decision making and problem solving.

**Temperature Example (Data → Information → Knowledge)**

**1. Data (Raw facts)**

- A list of temperature readings collected every hour:
  **22°C, 24°C, 26°C, 25°C, 27°C, 29°C**

- *Just numbers with no meaning yet.*

**2. Information (Processed data)**

- After organizing and analyzing the data:
  - **Minimum temperature:** 22°C
  - **Maximum temperature:** 29°C
  - **Average temperature:** 25.5°C

- *Data has been summarized and given context → now it has meaning.*

**3. Knowledge (Understanding & application)**

- From observing patterns over time:
  **"As temperature increases, energy usage also tends to increase (e.g., more air-conditioning is used)."**

- *We use experience + information to explain why something happens and make decisions.*

**Sales Example (Data → Information → Knowledge)**

**1. Data (Raw sales numbers)**

- Daily sales units recorded for a week:
  **120, 150, 130, 160, 140, 170, 180**

- *Just numerical values with no interpretation.*

**2. Information (Processed sales data)**

After organizing and analyzing the numbers:

- **Total sales for the week:** 1,050 units
- **Average daily sales:** 150 units
- **Highest sales day:** 180 units
- **Lowest sales day:** 120 units

- *Data has been summarized and now provides meaningful insights.*

**3. Knowledge (Understanding patterns & implications)**

From observing trends over time:

- **"Sales increase on weekends because more customers shop during that time."**
- **"Running promotions boosts sales by 20–30%."**

- *Information combined with experience helps explain **why** sales change and **how** to improve them.*

# Types of Data

## 1. Structured Data

Structured data is highly organized, stored in fixed rows and columns, and follows a predefined schema. Machines can easily search, query, process and analyze this type of data.
Example: Excel, CSV, Relational database

## 2. Semi-Structured Data

Semi-structured data does not follow a fixed schema, but still contains tags, or metadata that give some structure. It is easier than unstructured data to interpret and data is stored in hierarchical structures.
Example: JSON, XML, HTML Pages, Email messages, Geospatial data

### 3. Unstructured Data

Unstructured data has no predefined format, or schema. Machines cannot easily interpret the data and requires additional processing techniques like text cleaning, image processing, or feature extraction before it becomes useful.
Example: Text documents, Images, Videos, Audio, social media posts

## Types of data based on statistics

### 1. Qualitative Data

Qualitative data describes qualities or characteristics and it represent categories or labels rather than numbers.

**Subtypes of Qualitative Data**

**Nominal**: Categories without any order or ranking, example: Blood group, colours, places, etc.

**Ordinal**: Categories with a meaningful order or ranking, example: product ratings(1-5), customer reviews, etc.

### 2. Quantitative Data

Quantitative data consists of numerical values that can be measured or counted and it supports mathematical operations.

**Subtypes of Quantitative Data**

**Discrete**: Data that can take only specific integer values, it cannot take values between the discrete numbers. Example: number of students, count of product items, etc.

**Continuous**: Data that can take any value within a given range, can be fractional or decimal, it represent measurements. Example: height, weight, temperature, length, speed, etc.

# Data Mining

Data mining is the process of extracting useful patterns, trends, and insights from large datasets using techniques from statistics, and machine learning. It helps organizations analyze huge amounts of data to make decisions.

## Main Functionalities of Data Mining

**1. Classification**
It assigns each data item to a set of predefined categories or classes (e.g. classify emails as spam or not-spam).

**2. Regression**
It predicts numeric or continuous values based on input variables (e.g. future sales, prices).

**3. Clustering**
It groups similar data items together based on similarity, without using predefined labels, useful for discovering natural groupings or segments.

**4. Association Rule Learning**
It finds relationships or correlations between variables: identifies items/events that often occur together e.g. in retail data, customers who buy X often also buy Y.

**5. Outlier Detection**
It identifies unusual, or rare data points that are significantly different from the majority of data e.g. fraud detection, error detection, or spotting rare events.

**6. Trend Analysis**
It examines how data changes over time, detects trends, shifts or temporal patterns in datasets (e.g. time-series analysis).

## Interestingness of Patterns

When running a data-mining algorithm on a large dataset, we may often get many patterns. But not all patterns are meaningful or useful. A pattern is considered “interesting” only if it satisfies following criteria.

**Understandable**
The pattern should be easy to interpret and make sense to a human user not overly complex.

**Valid**
The pattern should hold not only on the sample data used to discover it, but also on new or test data (i.e. not be just a coincidence).

**Useful**
The pattern should provide insights or knowledge that can be used for decision-making, strategy, or improvement and not already known facts.

**Novel**
The pattern should bring something new and not something obvious or already known to the user. It should reveal unexpected or non-obvious relationships or insights.

# Data Preprocessing

Data preprocessing is the process of transforming raw data into a clean, structured, and analysis-ready form before applying data mining or machine learning algorithms. It deals with errors, inconsistencies, noise, missing values, and incompatible formats so that the data becomes reliable and interpretable.

In real-world datasets, data is messy, missing fields, duplicates, inconsistent formats and raw algorithms cannot work well without preprocessing.

## Importance of Data Preprocessing

Data preprocessing is essential because raw data often contains problems that can mislead analysis or degrade model performance. Key reasons we preprocess data include:

1. **Improves Data Quality:** Cleans errors, noise, and inconsistencies to make data accurate and consistent.
2. **Ensures Algorithm Compatibility**: Many data mining and ML algorithms require numerical, standardized input.
3. **Enhances Model Performance:** Clean, scaled, and properly formatted data leads to better accuracy and generalization.
4. **Reduces Bias and Variance Issues**: Handling missing values and outliers prevents models from being skewed.
5. **Improves Efficiency**: Cleaning and reducing data size speeds up computations and reduces resource usage.
6. **Facilitates Integration**: Prepares and standardizes data from multiple sources.

Improper preprocessing can lead to inaccurate or unreliable results, because algorithms rely heavily on the quality of input data.

## Data Preprocessing Techniques

- **Handling Missing Values**

Missing data occurs when some attribute values are absent in the dataset. If not handled properly, they can reduce accuracy, create bias and break algorithms that require complete data.

**Reasons Behind Missing Values in the Dataset**

Data can be missing from a dataset for several reasons and understanding the cause is important for selecting the most effective way to handle it. Common reasons for missing data include:

- **Technical issues:** Failed data collection or errors during data transmission.
- **Human errors:** Mistakes like incorrect data entry or oversights during data processing.

- **Privacy concerns:** Missing sensitive or personal information due to confidentiality policies.
- **Data processing issues:** Errors that occur during data preparation.

Common strategies to handle missing values:

**Mean imputation:** Replace missing numeric values with the mean of available values in the column. Useful when data is normally distributed.

**Median imputation:** Replace missing numeric values with the median of available values in the column. Useful when data is skewed and contains outliers.

**Mode imputation:** Replace missing categorical values with the most frequent category.

**Constant imputation:** Fill missing values with a fixed value like: "Unknown", "0", "Not Applicable"

- **Handling Outliers**

Outliers are extreme values that are significantly different from majority of data. They can distort summaries and models.

**IQR (Interquartile Range) Method:**

IQR is a method to detect outliers by looking at the spread of the middle 50% of data.
IQR is the range between the first quartile (Q1) and the third quartile (Q3).
It covers the middle 50% of the data.
It is used when data is skewed or non-normal distribution.

Steps:

1. Calculate Q1 (25th percentile)
2. Calculate Q3 (75th percentile)
3. Compute IQR = Q3 – Q1
4. Determine bounds:

$$\text{Lower} = Q1 - 1.5 \times IQR$$
$$\text{Upper} = Q3 + 1.5 \times IQR$$

Values outside this range are treated as outliers.

**Z-score method:**

This statistical method measures how many standard deviations a data point is from the mean.
The Z-score measures how far a data point is from the mean in terms of standard deviations.
Values with $|z| >$ threshold (e.g., > 3) as outliers.
It assumes data follows a normal distribution.

- **Encoding Categorical Values**

Most algorithms require **numerical input**, so categorical attributes must be encoded:

- **One-Hot Encoding:** Create binary columns for each category (e.g. Color: Red, Blue → columns Red, Blue). It is used for nominal data.
- **Label Encoding:** Assign numeric labels based on rank (e.g. Low=1, Medium=2, High=3). It is used for ordinal data.

- **Feature Scaling**

Feature values may be on very different numerical ranges, which can bias distance-based algorithms. Scaling ensures all features contribute equally.

- **Normalization:**
  It is also called as Min-Max Scaling.
  Scales values to the range [0 - 1].

- **Standardization:**
  It is also called as Z-score Scaling.
  Transforms data to mean = 0, standard deviation = 1.

- **Binning / Discretization**

Binning (also called **discretization**) groups continuous values into intervals (bins), helping simplify data, reduce noise, and improve robustness.

**Example of Binning**

Age (continuous):
12, 25, 33, 49 → Bins = {0–20, 21–40, 41–60}

**Types of Binning**

1. **Equal-width binning**
   All bins have the same interval size
2. **Equal-frequency binning**
   Each bin contains the same number of observations